What Is an Option?

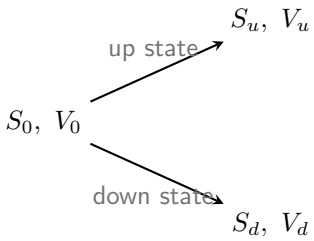
An option gives its holder the right, not the obligation, to transact in an underlying asset on specified terms.

- **Call:** right to buy at the strike price K.
- **Put:** right to sell at the strike price K.
- **Expiration** T**:** last date the right can be exercised.
- **Premium:** price paid today for the contract.

Central question

How much should someone pay today for a claim that provides a potentially uncertain payoff in the future?

An Arbitrary Financial Claim



Suppose a stock trades today at S_0. At time T, it can be S_u or S_d. A claim V written on the stock then pays V_u or V_d.

Key idea

We have not assumed V is a call or a put. We want a general method for pricing any financial claim.

Replicating the Financial Claim

Rather than price the claim directly, build a portfolio with exactly the same payoff in every state.

Replicating portfolio

Hold Δ shares of stock and borrow y dollars today at continuously compounded rate r.

ΔS_t is the stock position; ye^{rT} is the loan repayment at expiration.

$$V_u = \Delta S_u - ye^{rT},$$
$$V_d = \Delta S_d - ye^{rT}.$$

Why is the loan term the same?

The borrowing obligation is fixed in advance. It does not depend on whether the stock rises or falls.

Solving for Δ and the Loan

$$V_u - V_d = \Delta(S_u - S_d)$$

$$\therefore \qquad \Delta = \frac{V_u - V_d}{S_u - S_d}$$

$$y = e^{-rT}(\Delta S_d - V_d)$$

Δ sets the difference

It determines how many shares reproduce the gap between the two claim payoffs.

y sets the level

Borrowing shifts both portfolio payoffs by the same amount.

No-Arbitrage: The Law of One Price

Portfolio lemma

If two portfolios have identical future payoffs in every state, then an arbitrage-free market must assign them the same price today.

$$V_0 = \Delta S_0 - y$$

If the claim and its replica had different prices, an investor could buy the cheaper one and sell the more expensive one. Future cash flows cancel, leaving a riskless profit today.

Central insight

Price an uncertain payoff by constructing an equivalent payoff from assets whose prices are already known.

A One-Step Call Option Example

$S_0 = \$100$, $S_u = \$120$, $S_d = \$80$, $K = \$100$, and $r = 0$. Therefore,

$$V_u = \max(120 - 100, 0) = \$20, \qquad V_d = \max(80 - 100, 0) = \$0,$$

$$\Delta = \frac{20 - 0}{120 - 80} = 0.5, \qquad y = 0.5(80) - 0 = \$40,$$

$$V_0 = 0.5(100) - 40 = \boxed{\$10}.$$

Up state

Half a share is worth $\$60$. Repay $\$40$.
Remaining payoff: $\$20$.

Down state

Half a share is worth $\$40$. Repay $\$40$.
Remaining payoff: $\$0$.

The $\$10$ premium is paid today; $\$20$ or $\$0$ is paid at expiration.

From Replication to Risk-Neutral Pricing

Insert the replicating portfolio into $V_0 = \Delta S_0 - y$:

$$\begin{aligned} V_0 &= \Delta S_0 - e^{-rT}(\Delta S_d - V_d) \\ &= e^{-rT}\left[qV_u + (1-q)V_d\right], \end{aligned}$$

where

$$q = \frac{S_0 e^{rT} - S_d}{S_u - S_d}.$$

Interpretation

The price has the form of a discounted expected payoff. The weights q and $1-q$ are not an extra pricing assumption: they follow from the replicating portfolio.

What Are Risk-Neutral Probabilities?

$$S_0 = e^{-rT}\left[qS_u + (1-q)S_d\right]$$

The weights q and $1-q$ make the stock's expected return equal to the risk-free rate.

Actual probabilities

Describe beliefs about what will happen in the world.

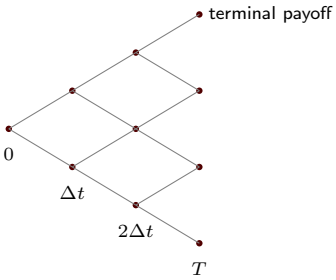
Risk-neutral probabilities

Are pricing weights consistent with traded-asset prices and no-arbitrage.

No-arbitrage condition

$S_d < S_0 e^{rT} < S_u$ ensures $0 < q < 1$. These weights arise from replication, not from forecasting the next stock move.

Extending to Multiple Time Steps



At every terminal node, the contract tells us the payoff. Then work backward:

$$V_t = e^{-r\Delta t}\left[qV^u_{t+\Delta t} + (1-q)V^d_{t+\Delta t}\right].$$

1. Calculate terminal payoffs.
2. Weight and discount adjacent values.
3. Repeat until time zero.

Terminal nodes are model-generated scenarios, not forecasts of realized prices.

From Binomial Trees to Black–Scholes

Discrete binomial model

Stock prices move up or down at individual time steps.

$\rightarrow$

Continuous-time model

Stock prices evolve continuously under a stochastic process.

More steps, smaller Δt

Under an appropriate parameterization, the binomial price converges to the Black–Scholes–Merton price as the number of steps grows.

What does not change

Replicate the claim, eliminate hedgeable risk, and use no-arbitrage to determine its price.

Black–Scholes–Merton: Equation and Price

Continuous-time delta hedging and no-arbitrage lead to

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

For a European call with no dividends,

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2),$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Terms in the PDE

Time sensitivity, stock sensitivity, curvature, volatility, and risk-free financing.

Assumptions include frictionless trading, constant r and σ, continuous hedging, lognormal dynamics, and no dividends.

What Does Black–Scholes Actually Tell Us?

It calculates

A theoretical arbitrage-free option premium under its assumptions.

It does not calculate

A prediction of the realized stock price at expiration.

$$C_T = \max(S_T - K, 0),$$
$$P_T = \max(K - S_T, 0).$$

Terminal payoff depends on the unknown S_T. A traded market premium can differ from the model price.

Keep the terms separate

The premium is the upfront price of the contract, not a fee added to its theoretical value.

The Greeks: Sensitivities of Option Value

Delta: $\frac{\partial V}{\partial S}$

Sensitivity to stock price; locally, the shares needed to hedge the option.

Gamma: $\frac{\partial^2 V}{\partial S^2}$

Sensitivity of Delta to the stock price.

Theta

Sensitivity to time; its sign depends on whether time is calendar time or time to expiration.

Vega: $\frac{\partial V}{\partial \sigma}$

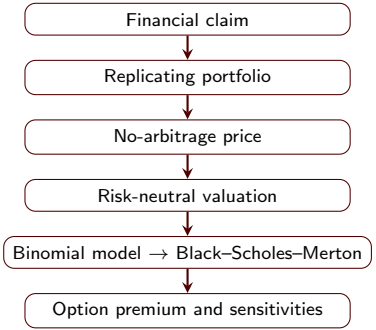
Sensitivity to volatility.

Rho: $\frac{\partial V}{\partial r}$

Sensitivity to the interest rate.

Connection: the Δ from the one-step replicating portfolio becomes a local, continuously adjusted hedge ratio.

Final Takeaway



Central message

Options pricing is not fundamentally about predicting where a stock will go. It is about determining what a financial claim should cost today, given its possible future payoffs and the prices of tradable assets.